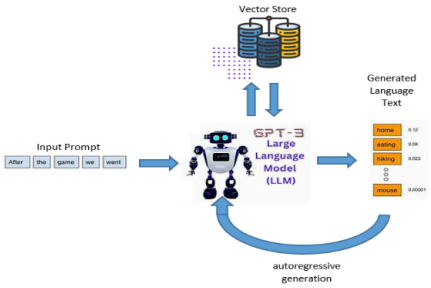
ITC C508

Lecture 4 : Introduction to LLM Ops and RAG Concepts



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About AI Assistants



- AI assistants, also known as chatbots, are computer programs designed to simulate conversations with human users.
- They are powered by advanced Natural Language Processing (NLP) algorithms to generate responses.
- Utilizing GPT-3 and other Large Language Models (LLMs) technologies, these programs have the ability to understand and interpret human language, allowing them to answer questions and perform various tasks
- They have found applications in various fields, including customer service, education, and entertainment.
- They are becoming increasingly sophisticated and are changing the way people interact with technology.

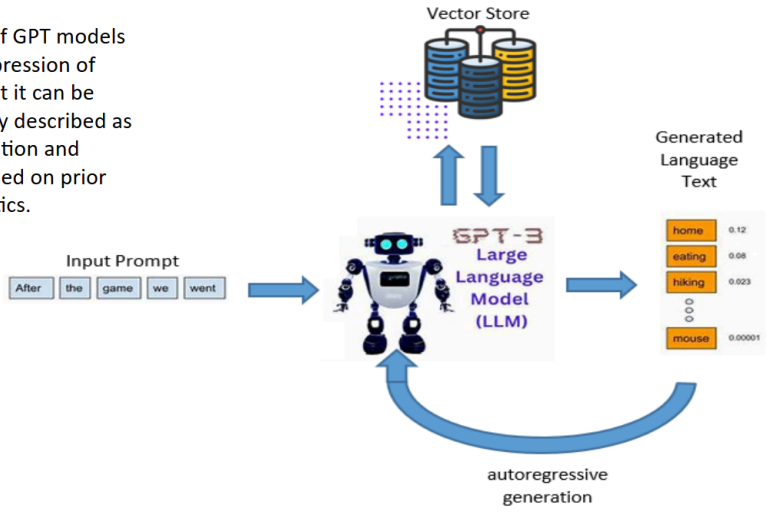
A Large Language Model : GPT



- GPT (for Generative Pretrained Transformer) is an advanced Large Language Model developed by OpenAI and corresponds to the right part of the Transformers architecture.
- It is capable of generating human-like text and performing various natural language processing tasks such as text completion, translation, and summarization.
- GPT is a next-token predictor, meaning it generates the next token in a sequence based on probabilities, starting from an initial prompt. Once a token is generated, it is stored in the "memory" along with the initial prompt.
- Consequently, if you provide GPT with a question in addition to a specific context, it can generate an accurate answer.
- However, the probabilistic approach employed by GPT *does not guarantee the accuracy of the information provided*. This phenomenon is referred to as model **hallucination**.

GPT models are not really Intelligent

The response of GPT models can give an impression of intelligence, but it can be more accurately described as pattern recognition and completion based on prior data and statistics.



Large Language Model Operations (LLMOps) refers to the practices and tools for developing, deploying, and maintaining large language models in production environments.



The Rise of LLMOps



- Large Language Models (LLMs) like GPT-3, BERT, and RoBERTa have revolutionized Natural Language Processing (NLP).
- They can understand and generate human language, but managing them is a complex task.
- These models require significant computational resources and continuous monitoring to maintain performance.
- This is the challenge that **Large Language Model Operations (LLMOps)** was created to solve.
- The term LLMOps gained popularity with the release of ChatGPT in 2022.

MLOps vs LLMOps

While both MLOps and LLMOps are essential for AI development, they address different challenges. The key distinctions are summarized below:

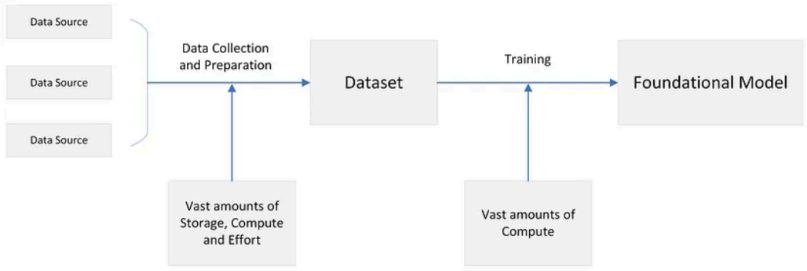
Feature	MLOps (Machine Learning Operations)	LLMOps (Large Language Model Operations)
Scope	General practices for the entire machine learning model lifecycle.	A specialized subset of MLOps, focused exclusively on Large Language Models (LLMs).
Primary Goal	To streamline and automate the development, deployment, and maintenance of ML models.	To address the unique challenges of developing and deploying LLMs in production environments.
Key Challenges Addressed	Version control, reproducibility, data drift, and scalability for ML models.	Model size, high computational cost, prompt management, and ensuring content safety and ethical guidelines.

Components of LLMOps

1. **Data Collection and Preparation**
2. **Model Development**
3. **Prompt Engineering, RAG and Model Fine-tuning**
4. **Model Deployment**
5. **Observability**
6. **Reinforcement Learning from Human Feedback (RLHF)**

1. Data Collection and Preparation

- Data collection and preparation are critical steps for training or fine-tuning a Large Language Model (LLM).
- When training an LLM from scratch, both a vast quantity and high quality of data are required to produce an effective model.



Model Development

- **The Challenge**: Most organizations lack the massive compute and storage to train LLMs from scratch.
- **The Solution**: Organizations start by consuming pre-trained LLMs via APIs or by directly using the models.
- **How to Choose the Right LLM**: The decision depends on several key factors:
  - Performance:
    - Use established leaderboards to compare models, such as:
    - Hugging Face's 'Open LLM Leaderboard
    - 'Stanford's 'HELM (Holistic Evaluation of Language Models)'
  - Business Constraints: Consider practical factors like:
    - Cost
    - Token limits
    - Model availability

Prompt Engineering, RAG and Model Fine-tuning

- In the evolving landscape of artificial intelligence, simply using a Large Language Model (LLM) is no longer enough to achieve optimal results for specific tasks.
- To unlock their full potential and tailor them to unique needs, practitioners are increasingly relying on a suite of advanced techniques.
  - **Prompt Engineering:** The art and science of crafting effective inputs to guide LLMs toward desired outputs.
  - **Retrieval-Augmented Generation (RAG):** A powerful approach that grounds an LLM's responses in external, verifiable data to improve relevance and reduce inaccuracies.
  - **Model Fine-tuning:** The process of taking a pre-trained LLM and training it on a smaller, task-specific dataset to adapt its knowledge and behavior.

Prompt Engineering

- **Definition:** The iterative process of refining and optimizing inputs (prompts) to guide an LLM toward a desired, high-quality output.
- **Techniques (Prompting with Examples):**
  1. Zero-shot: Providing no examples in the prompt.
  2. Single-shot: Including one example of the desired output format.
  3. Few-shot: Providing multiple examples to guide the LLM's response.
- **Best Practices:**
  - Experiment with different prompt templates and phrasing.
  - Keep track of prompt variations and other variables, such as the model's temperature setting, to find the optimal combination for your task.
  - The goal is to find the most effective prompt that consistently yields the best output.

Retrieval-Augmented Generation (RAG)

- **Purpose:** A method to enhance LLM performance by providing models with relevant, external information (context) to generate more accurate and up-to-date responses.
- **How it Works:**
  1. The LLM is given a query and asked to address it using the provided context.
  2. To ensure the context is relevant, Vector Stores are used. These can be databases (e.g., Pinecone, ChromaDB) or libraries (e.g., FAISS).
  3. Vector stores store data as embeddings and retrieve the most relevant information based on the user's query.
  4. Both the original prompt and the retrieved context are then passed to the LLM to generate a final, informed output.

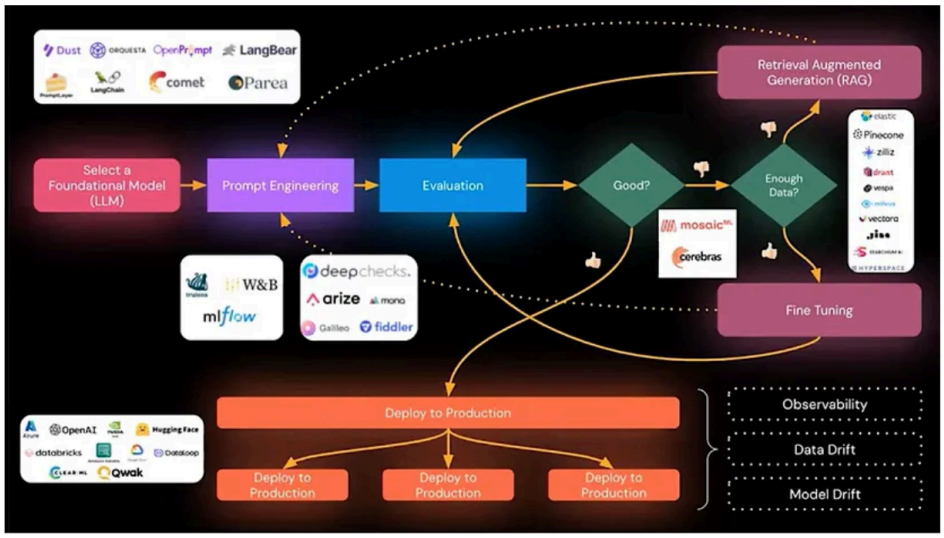
Model Fine-tuning

- **Purpose:** A powerful technique used when Prompt Engineering and RAG are insufficient. It involves training an LLM on a smaller, task-specific dataset to adapt its internal knowledge and behavior.
- **Key Considerations:**
  - **Cost**: Fine-tuning is computationally expensive due to the significant resources required for training and data preparation.
  - **Black-Box Problem**: Even after fine-tuning, LLMs can produce unpredictable outputs, as their complex internal workings are difficult to fully understand and control.
- **Best Practices**
  - Versioning is critical. Track different versions of your fine-tuned model, similar to MLOps practices, to manage updates and rollbacks.
- **The Next Step:**
  - If fine-tuning does not yield satisfactory results, the ultimate step is to train a custom model from scratch using your own data.

LLM Deployment

- **The Goal:** After an LLM has been developed from scratch, fine-tuned, or selected as an open-source model, the next step is to deploy it for production use.
- **Primary Challenges:**
  - **Model Size:** LLMs are typically very large, which can complicate deployment..
  - **Compute Power:** They require significant computational resources to run effectively.
- **Critical Performance Metrics:**
  - **Latency**: A key metric that measures the time it takes for the model to generate a response.
  - **Completion Length**: The length of the LLM's generated output can have a direct and significant impact on latency. It is an important factor to consider during deployment.

LLM Life Cycle



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